Transformer MLP Transplants Depend on the Receiving State

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Abstract

Activation patching and steering move internal updates between forward passes and often treat a moved update as if it carried a portable effect. Ordinary transplants cannot test this, because the update and the state that receives it change together. We isolate the receiver with a crossed-receiver design: a natural transformer MLP displacement $\Delta u = u_j - u_i$ is computed once, frozen, and physically added to many natural receiver states of Qwen3-8B and Mistral-7B; its exact next-token distributional effect is measured as a KL divergence. The same displacement is mild in some receivers and disruptive in others; across 31 receivers its effect typically spans more than three orders of magnitude. An established receiver-local Fisher score ranks these full-dose effects consistently, with median within-displacement Spearman correlation 0.858–0.928 across all six model–bucket replications; the corresponding unweighted response reaches at most 0.568. The association remains after removing generic displacement strength and receiver fragility, with residual correlation 0.50–0.68. The receiver also changes the direction of the effect: footprints of one displacement diverge across receivers, yet receiver-specific low-rank maps fitted on other displacements predict held-out footprints better than a global map. The effect of an internal update depends on where it is written, in both size and direction.[1]

1 Introduction

Activation patching, path patching, and causal tracing move hidden activations between forward passes to localize the components that drive a behavior [Vig et al., 2020, Geiger et al., 2021, Goldowsky-Dill et al., 2023, Zhang and Nanda, 2024]; activation addition and function vectors move them to control behavior [Turner et al., 2023, Todd et al., 2024]. Both uses read a moved activation as carrying an effect. That reading is only as portable as the downstream network allows, because the remaining layers process the moved activation together with the state that receives it.

Transformer MLPs make the question concrete. At layer ℓ, a host context i and a donor context j produce MLP outputs u_i and u_j. Replacing u_i by u_j is exactly adding the natural displacement $\Delta u_{ij} = u_j - u_i$ to the host's residual stream (Figure 1a). In an ordinary transplant, every host receives a different displacement, so the update and its destination change together. Ordinary transplants can show that host-side quantities are associated with transplant effects (Appendix D.1), but not whether the receiving state changes what a fixed update does.

In this paper, we isolate the receiver. In the crossed-receiver design (Figure 1b, Algorithm 1), a natural displacement is computed once, frozen, and physically added, with unchanged direction and norm, to many other natural receiver states. Because the update is identical across receivers, any variation of its exact effect is caused by the receiving state and its interaction with the update. We

[1] Code and reproducibility: github.com/mnehan67/neurips-2026

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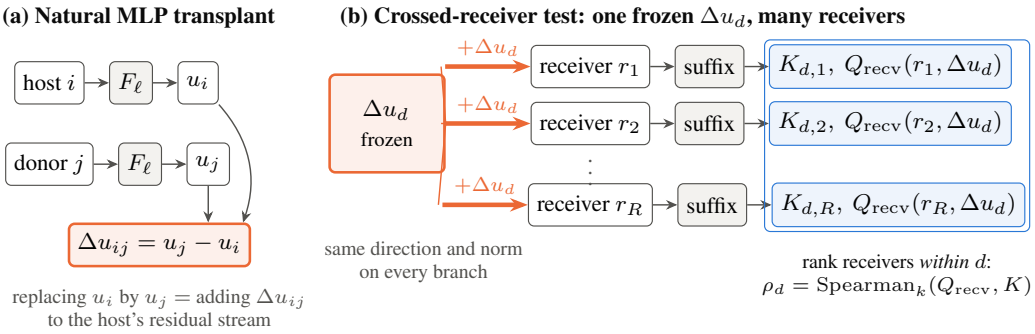


Figure 1: **The crossed-receiver design holds the update fixed and changes only its destination.** (a) Replacing the host MLP output u_i by the donor output u_j adds the natural displacement $\Delta u_{ij} = u_j - u_i$ to the host. (b) In the crossed-receiver test, one natural displacement Δu_d is computed once, frozen, and added unchanged to many natural receiver states $r_1, \ldots, r_R$. For each receiver we measure the exact effect $K_{d,k}$ and the receiver-local score $Q_{\text{recv}}(r_k, \Delta u_d)$, and we rank receivers within the fixed displacement (Algorithm 1).

explain this variation in two steps. For its *size*, we use the receiver-local Fisher score, the standard second-order pullback of output KL through the downstream network [Martens, 2020, Park et al., 2026, Wang and Zhao, 2026], as a diagnostic; we do not claim a new geometry. For its *direction*, we trace the causal footprint of the displacement through every downstream layer and test whether receiver-specific structure in it generalizes to displacements never used for fitting.

What the design adds beyond nonlinearity. A nonlinear network is state dependent in general, so receiver dependence alone is not our claim. We ask how large and how structured it is for natural MLP updates at full dose, with three tests that do not follow from nonlinearity: the primary statistic ranks receivers while the physical update is held fixed; a residual analysis removes generic update strength and receiver fragility before asking whether the Fisher score tracks the pair-specific remainder; and a held-out map test asks whether the receiver-dependent change of direction transfers to unseen displacements, which tests for reusable receiver-specific structure beyond generic state dependence.

Our contributions are as follows (related work is discussed in Appendix A):

- **Crossed-receiver design.** We hold a natural MLP displacement physically fixed and vary only its receiver (Algorithm 1). In Qwen3-8B and Mistral-7B, across three disjoint data buckets and four depths, the exact full-dose effect of a fixed displacement varies strongly and systematically with the receiver (Section 5.2).
- **Receiver-side explanation of effect size.** The established receiver-local Fisher score ranks receivers with median within-displacement $\rho = 0.858$–0.928, ahead of matched-magnitude, attribution-patching, relative-norm, receiver-norm, and alignment baselines (Table 2), and retains a residual association of 0.50–0.68 with pair-specific variation (Table 3).
- **Receiver-conditioned transformation of direction.** The same displacement acquires different downstream causal directions in different receivers, and receiver-specific rank-16 maps predict a reproducible component of this transformation for held-out displacements in both model families (Figure 3).

2 Problem Setup

Fix a frozen decoder-only transformer, an MLP layer ℓ, and contexts of 64 tokens. All interventions act on the MLP output at the final token, and no model parameter is changed.

Natural displacements. Let F_ℓ be the MLP at layer ℓ and let m_c be its input at the final token of context c. A host i and a donor j produce

$$u_i = F_\ell(m_i), \qquad u_j = F_\ell(m_j), \qquad \Delta u_{ij} = u_j - u_i. \tag{1}$$

It is *natural*: both endpoints are MLP outputs the model actually produces.

Algorithm 1 Crossed-receiver test for one natural displacement

Require: frozen model; layer ℓ; host i; donor j; receivers $k_1, \ldots, k_R$ (natural contexts)
$\Delta u_d \leftarrow F_\ell(m_j) - F_\ell(m_i)$ ▷ natural displacement, computed once
freeze Δu_d ▷ identical direction and norm for every receiver
for $k \in \{k_1, \ldots, k_R\}$ **do**
  run receiver k natively; record r_k and p_k
  run receiver k with $r_k \leftarrow r_k + \Delta u_d$; record $p_{k,+\Delta u_d}$
  $K_{d,k} \leftarrow D_{\mathrm{KL}}(p_k \,\|\, p_{k,+\Delta u_d})$ ▷ exact full-dose effect
  record predictors at $(r_k, \Delta u_d)$: Q_{recv}, q_{p2}, attribution, norm, and alignment scores
  record causal footprints $c^{(m)}_{d,k} = h_m(k; +\Delta u_d) - h_m(k)$ ▷ control panel only
end for
$\rho_d \leftarrow \mathrm{Spearman}_{k \neq i}\big(\text{predictor}_{d,k},\, K_{d,k}\big)$ ▷ update fixed inside ρ_d
return ρ_d; the primary statistic is $\mathrm{median}_d\, \rho_d$ over displacements

Receivers and the exact effect. For any context k, let r_k be the residual state immediately after the layer-ℓ MLP at the final token, and let p_k be the native next-token distribution. Writing a displacement Δu into receiver k replaces r_k by $r_k + \Delta u$ (equivalently, the MLP output u_k by $u_k + \Delta u$) and leaves the rest of the forward pass unchanged. With $p_{k,+\Delta u}$ the resulting next-token distribution, the exact next-token distributional effect (for short, the exact effect) is

$$K_k(\Delta u) = D_{\mathrm{KL}}\big(p_k \,\big\|\, p_{k,+\Delta u}\big)\,. \tag{2}$$

The forward KL from the native distribution, over the full vocabulary, measures how far the physically applied intervention moves the model from its unpatched behavior. An ordinary transplant is the special case $K_{i \leftarrow j} = K_i(\Delta u_{ij})$; a self-patch recovers the native output to numerical precision.

Goal. Our goal is to determine whether, for a fixed natural displacement Δu_d, the effect $K_k(\Delta u_d)$ depends on the receiver k, how strongly it does so, and whether receiver-side quantities explain both the size and the direction of the dependence.

3 The Crossed-Receiver Design

We first present the protocol and its statistics; the receiver-side quantities it records are defined in Section 4.

3.1 Protocol

Each crossed experiment samples natural displacements $\Delta u_d = u_{j_d} - u_{i_d}$ from host–donor pairs and crosses each with a set of natural receivers, never rescaling or re-orienting it (Algorithm 1). The original host is kept as an anchor row; the primary statistic uses only alternative receivers $k \neq i_d$.

3.2 Three correlations, one primary statistic

The protocol yields displacement-by-receiver matrices of exact effects $K_{d,k}$ and predictor values $X_{d,k}$. We summarize their association in three ways that answer three different questions (formal definitions in Appendix B.4).

1. **Pooled correlation** $\rho_{\mathrm{pool}} = \mathrm{Spearman}(X_{d,k}, K_{d,k})$ over all rows. It is descriptive and mixes update, receiver, and pair-specific variation.
2. **Within-displacement correlation (primary)** $\mathrm{median}_d\ \rho_d$, where ρ_d correlates $X_{d,k}$ and $K_{d,k}$ across the receivers of one fixed displacement. It asks whether X ranks receivers correctly *while the physical update is held fixed*.
3. **Residual-rank correlation.** Rank-transform X and K, remove additive displacement and receiver main effects from each by alternating demeaning, and correlate the residuals. It asks whether X still captures pair-specific structure *after generic displacement strength and receiver fragility are removed*. It was introduced after inspecting the crossed data and is reported as post hoc.

These are not competing estimates of one quantity. In the six crossed matrices, receiver identity alone explains 0.70–0.80 of the rank variance of exact KL, displacement identity 0.04–0.12, and 0.18–0.25 remains as non-additive residual variation (Table 4). A high within-displacement correlation thus largely means that a predictor identifies generically fragile receivers; the residual correlation asks whether it also tracks the smaller non-additive part. Values near 0.9 for the first question and 0.5–0.68 for the second are therefore consistent. With one effect per cell, we do not interpret the remainder as a separately identified interaction. Every Fisher–KL correlation in the paper is one of these types on a specific panel; Table 5 lists them side by side.

What the design identifies. Within a displacement the intervention vector is physically identical across receivers, so differences in $K_{d,k}$ are caused by the receiving state and its interaction with the update, for the given model and layer. The design does not identify which mechanism (attention, normalization, or MLP gating) produces the dependence, nor whether an MLP update represents a semantic function; our endpoints are next-token behavior and causal-effect geometry.

4 Receiver-Side Measurements

4.1 Receiver-local sensitivity

Let G_ℓ map the post-MLP residual state at layer ℓ to the final logits, with the rest of the context held fixed. A small movement of receiver r along Δu changes the logits by $\Delta z = JG_\ell(r)\,\Delta u$. With p the receiver's native next-token distribution and $C_p = \mathrm{diag}(p) - pp^\top$, the receiver-local score is

$$Q_{\mathrm{recv}}(r, \Delta u) = \tfrac{1}{2}\left(JG_\ell(r)\Delta u\right)^\top C_p \left(JG_\ell(r)\Delta u\right). \tag{3}$$

This is the standard second-order pullback of output KL, $D_{\mathrm{KL}}(p_r \,\|\, p_{r+\epsilon\Delta u}) = \epsilon^2\, Q_{\mathrm{recv}}(r, \Delta u) + O(\epsilon^3)$ [Martens, 2020, Park et al., 2026, Wang and Zhao, 2026] (Appendix B.1); we use it as a diagnostic rather than claim a new geometry. The empirical question is whether this local quantity, evaluated at each receiver, ranks the *full-dose* effect ($\epsilon = 1$) of a fixed natural displacement.

Remark 1 (Standard variance reading). $(JG_\ell\Delta u)^\top C_p\,(JG_\ell\Delta u) = \mathrm{Var}_{y\sim p}\big[(JG_\ell\Delta u)_y\big]$, so Q_{recv} is half the variance of the logit change under the receiver's own next-token distribution. A uniform logit shift costs nothing, and changes on tokens the receiver considers unlikely cost little. The matched baseline $q_{p2} = \frac{1}{2}\|JG_\ell(r)\Delta u\|_2^2$ uses the same response without this weighting, isolating the contribution of the receiver's output distribution.

The crossed experiments estimate $JG_\ell(r)\Delta u$ by centered finite differences ($\epsilon = 0.075$) on FP16 model evaluations. An exact FP32 downstream-suffix JVP audit on 16 control-panel cases per model shows negligible truncation error (median cosine 1.0000; $\rho = 1.000$ between FP32 finite-difference and exact Fisher scores), whereas FP16 scores for the same cases agree less closely ($\rho = 0.785$, 0.544; Appendix D.12). We therefore use the FP16 score as a ranking diagnostic: the primary crossed result is an empirical association between this computed score and the independently measured full-dose KL, and it does not require the score to be a numerically exact FP32 Fisher quadratic. The directional results (Sections 5.5 and 5.6) do not use the score.

Alternative predictors. We compare Q_{recv} with q_{p2}; with first-order attribution of the native top-token log-probability (AtP-top) and of the native top1–top2 logit margin (AtP-margin), in the spirit of attribution patching [Kramár et al., 2024]; with the relative displacement norm $\|\Delta u\|/\|r\|$; with the receiver norm $\|r\|$; and with the alignment $|\cos(\Delta u, r)|$. We also run a physical control in which every natural displacement is rescaled to a common norm before being applied (Appendix C.5).

Remark 2 (Why KL is not attributed directly). With p_α the distribution after adding $\alpha\Delta u$, $\frac{d}{d\alpha} D_{\mathrm{KL}}(p \,\|\, p_\alpha)\big|_{\alpha=0} = 0$ (Appendix B.3), so first-order attribution of KL is identically zero and AtP uses the non-stationary targets above, in absolute value.

4.2 Following the causal effect downstream

KL measures how much the final distribution changes, but not the direction of the hidden-state effect. For displacement d with original host i and receiver k, the causal footprint at downstream layer m is

$$c_{d,k}^{(m)} = h_m(k; +\Delta u_d) - h_m(k), \tag{4}$$

Table 1: **Experimental panels.** All panels use frozen weights, 64-token C4 validation contexts, and final-token MLP interventions. "Rows" counts interventions per depth, per bucket, or per Pythia run (256 hosts); a depth count of 1 means the primary depth (layer 2 for Pythia). Crossed receiver counts include the original host as an anchor row; the within-displacement statistic uses the 15 or 31 alternative receivers.

Panel	Design	Rows	Depths	Used for
Ordinary	all directed pairs, 64 contexts	4032	4	residual; wrong-state; dose; horizon
Large ordinary	Mistral, 256 contexts	65,280	1	scale; permutation test
Crossed B1	512 $\Delta u \times 16$ receivers	8192	4	primary statistic; depth
Crossed B2, B3	512 $\Delta u \times 32$ receivers	16,384	1	replication
Control	96 $\Delta u \times 8$ receivers	768	1	baselines; footprints; maps
Pythia-70M ($\times 6$)	ordinary, 5-NN donors	256	1	independent pretraining

the difference between patched and native final-token hidden states, with $c^{(m)}_{d,i}$ defined analogously at the original host. Both footprints equal Δu_d at the intervention site. We summarize how they separate with the directional fidelity $F^{(m)}_{d,k} = \cos(c^{(m)}_{d,i}, c^{(m)}_{d,k})$, the gain $A^{(m)}_{d,k} = \langle c^{(m)}_{d,i}, c^{(m)}_{d,k} \rangle / \|c^{(m)}_{d,i}\|^2_2$, and the rewrite residual $E^{(m)}_{d,k} = c^{(m)}_{d,k} - A^{(m)}_{d,k} c^{(m)}_{d,i}$. Together they distinguish an effect that is preserved but attenuated or amplified ($F \approx 1$, $A \neq 1$) from one whose direction is rewritten (small F, large E).

Held-out receiver maps. To test whether the receiver-dependent transformation has reusable structure, we fit at each downstream layer a low-rank map $T^{(m)}_k$ predicting $c^{(m)}_{d,k}$ from $c^{(m)}_{d,i}$. Displacements are split once, before analysis, into 75% training and 25% held out; a rank-16 PCA basis is fit on training displacements only, and ridge maps are fit in that subspace. We compare five predictors of the held-out footprint: the *identity* ($c_k \approx c_i$), a *gain-only* scalar per receiver, one *global map* shared by all receivers, the *receiver-specific map*, and a *wrong-receiver map* taken from another receiver by a fixed cyclic permutation. If receiver identity carried no reusable structure beyond a shared transformation, receiver-specific maps should not systematically outperform a global map on held-out displacements; under a pure change of scale, gain only would suffice. We score cosine and normalized RMSE on held-out displacements, with the held-out displacements as independent units after averaging receivers (Appendix B.5).

5 Experiments

We apply the design to two 7–8B models, then test the size explanation against baselines and pair-specific structure, and finally the direction of the effect and its generalization.

5.1 Setup

Models. We use Qwen3-8B [Yang et al., 2025] (36 layers) and Mistral-7B-v0.3 [Jiang et al., 2023] (32 layers) with frozen FP16 weights. The four probed depths are Qwen layers 7, 14, 21, 26 and Mistral layers 6, 12, 19, 23; the primary depths, at about 40% of each network, are Qwen layer 14 and Mistral layer 12. Six independently pretrained Pythia-70M runs [Biderman et al., 2023, van der Wal et al., 2025] provide a separate ordinary-transplant replication (Section 5.7).

Data. Contexts are deterministic samples from the C4 validation split [Raffel et al., 2020], selected by a SHA-256 hash rule before model-specific tokenization. Each context has 64 tokens, and every intervention and measurement uses the final token. The three crossed buckets B1–B3 are assigned by distinct hash residues and are therefore disjoint.

Panels. Table 1 lists the panels. The crossed buckets carry the primary result. The control panel is a separate deterministic 96×8 panel, sampled with a different seed and selection rule; it stores layerwise hidden-state differences for the baselines, footprints, and held-out maps, and its 96 displacements are split once into 72 training and 24 held-out. The primary statistic and baselines are those of Sections 3.2 and 4.1.

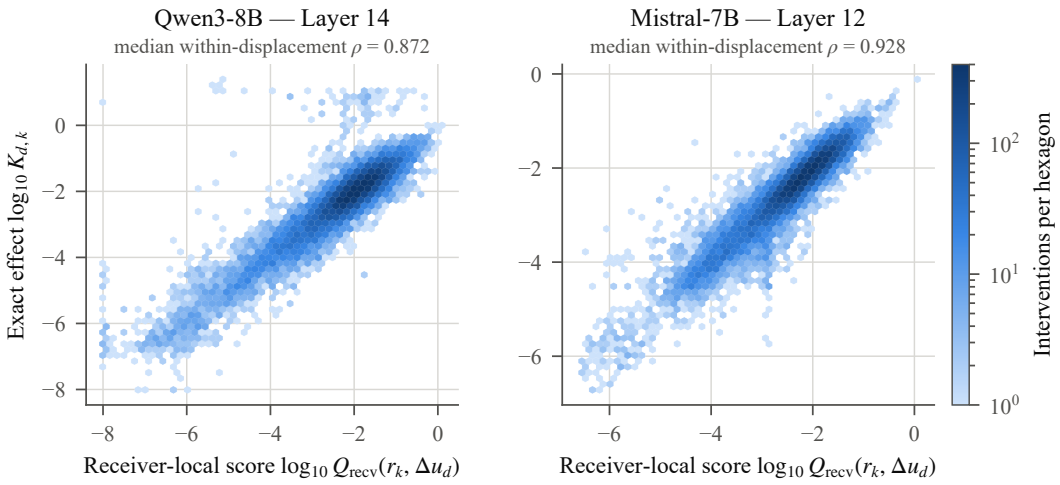


Figure 2: **Receiver-local sensitivity ranks the effects of a fixed displacement across receivers.** Bucket B2, primary depth: 512 frozen natural displacements, each applied to 31 alternative receivers (15,872 interventions per model); both axes $\log_{10}$. Headers give the primary statistic. Other buckets and depths: Tables 3 and 6.

Table 2: **Alternative predictors of receiver ordering.** Median within-displacement Spearman correlation (↑) with exact full-dose KL on the 96×8 control panel; bold marks the best per model. With 8 receivers per displacement, medians are coarser and lower than in the 32-receiver buckets (Table 5); pooled and residual versions are in Table 8.

Model	Fisher Q_{recv}	AtP-top	AtP-margin	q_{p2}	$\|\Delta u\|/\|r\|$	$\|r\|$	$\|\cos(\Delta u, r)\|$
Qwen3-8B	**0.738**	0.500	0.429	0.226	0.071	−0.071	−0.012
Mistral-7B	**0.905**	0.750	0.179	0.060	0.000	0.000	0.048

Implementation details. We use Hugging Face Transformers and shard the large models with Accelerate across two NVIDIA T4 GPUs (16 GB each). No model is trained or fine-tuned. Seeds, context identifiers, and the split are fixed before analysis. Protocols, compute, and frozen outputs are in Appendix C and at https://anonymous.4open.science/r/newinml-7402/.

5.2 The same displacement has different effects in different receivers

A fixed natural displacement does not have one effect. With 31 alternative receivers (buckets B2 and B3), the largest and smallest exact effects of the same displacement differ by a median factor of $10^{4.4}$–$10^{4.8}$ in Qwen3-8B and $10^{3.3}$–$10^{3.6}$ in Mistral-7B, and by at least a factor of 100 for 99.0–100% of displacements (Appendix D.5).

This variation is ordered. The receiver-local score ranks receivers with median within-displacement correlation $0.871/0.872/0.858$ in Qwen3-8B buckets B1/B2/B3 and $0.911/0.928/0.928$ in Mistral-7B (Figure 2), and the correlation is positive for 99.4–99.8% of Qwen displacements and every sampled Mistral displacement. The unweighted response q_{p2}, computed from the same finite differences, reaches only 0.340–0.355 in Qwen and 0.510–0.568 in Mistral, so weighting by the receiver's own next-token distribution (Remark 1) accounts for much of the ranking.

5.3 Scale, attribution, and alignment baselines do not explain the ordering

On the control panel (Table 2), the Fisher score is the strongest predictor in both models. Attribution of the native top-token log-probability is the closest competitor (0.500 in Qwen3-8B, 0.750 in Mistral-7B); the relative displacement norm, receiver norm, and displacement–state alignment are near zero. A physical control gives the sharpest scale check: rescaling every natural displacement to a common norm before applying it leaves the receiver ordering of exact KL unchanged (median within-displacement $\rho = 1.000$ between original and equal-norm orderings in both models). Since a

Table 3: **Within-displacement and pair-specific association in the six crossed replications.** "Within" is the primary statistic. "Residual" is the post-hoc rank correlation after removing additive displacement and receiver effects (Section 3.2). "Non-additive" is the share of rank-KL variance not explained by the additive displacement-plus-receiver model (Table 4). Higher is better for all correlation columns.

		Within-displacement ρ		Residual ρ		
Model	Bucket	Fisher Q_{recv}	q_{p2}	Fisher Q_{recv}	q_{p2}	Non-additive
Qwen3-8B	B1	0.871	0.355	0.504	0.284	0.211
Qwen3-8B	B2	0.872	0.340	0.504	0.292	0.221
Qwen3-8B	B3	0.858	0.352	0.520	0.317	0.246
Mistral-7B	B1	0.911	0.529	0.664	0.411	0.196
Mistral-7B	B2	0.928	0.510	0.675	0.442	0.185
Mistral-7B	B3	0.928	0.568	0.658	0.414	0.180

displacement's scale factor is shared by all its receivers, displacement norm does not produce the receiver ordering.

5.4 Pair-specific structure remains after removing generic fragility

A high within-displacement correlation could reflect generic receiver fragility alone. Table 3 separates the two: 18–25% of the rank variance of exact KL is non-additive in every crossed matrix, and after removing additive displacement and receiver effects the Fisher score's residual correlation is 0.504–0.520 in Qwen3-8B and 0.658–0.675 in Mistral-7B, above q_{p2} in all six replications (0.284–0.442). The score therefore tracks pair-specific variation, not only fragility; the analysis is post hoc and descriptive.

5.5 The receiver changes the direction of the causal effect

The scalar results leave open whether receivers merely amplify or attenuate a fixed update. On the control panel, the footprints of one displacement at the original host and at a receiver start identical and then separate quickly: median directional fidelity falls from 1 to 0.030 at Qwen3-8B's final layer and 0.037 at Mistral-7B's, and the median gain decays with it (Figure 4). The receiver's final footprint is nearly orthogonal to the host's, so the receiver rewrites the direction of the effect rather than rescaling a shared one.

5.6 A receiver-specific transformation generalizes to held-out displacements

Figure 3 shows that part of this rewriting is systematic. At the final layer, receiver-specific maps reach mean held-out cosine 0.316 in Qwen3-8B and 0.241 in Mistral-7B, against 0.086 and 0.109 for the global map and -0.023 and 0.028 for wrong-receiver maps. Bootstrapping the 24 held-out displacements, the advantage over the global map is $+0.230$ (95% CI $[0.205, 0.256]$) in Qwen3-8B and $+0.132$ ($[0.108, 0.155]$) in Mistral-7B, and over wrong-receiver maps $+0.340$ ($[0.281, 0.400]$) and $+0.213$ ($[0.180, 0.244]$): the predictive structure belongs to the specific receiver. Exact vector reconstruction improves much less (normalized RMSE 0.945 versus 1.005 for the global map in Qwen3-8B, 0.986 versus 0.999 in Mistral-7B; Table 9). The transformation thus contains a reproducible receiver-specific *component*, while substantial update-specific variation remains.

5.7 Robustness

Depth and scale. In the B1 depth sweep, the median within-displacement correlation ranges from 0.857 to 0.954 across the four Qwen depths and from 0.854 to 0.957 across the four Mistral depths (Table 6). A 65,280-pair Mistral matrix of ordinary transplants gives Fisher–KL correlation 0.911 and host–donor residual correlation 0.660 (q_{p2}: 0.415), with joint context-label permutation $p = 0.0005$, the smallest value attainable with 2000 permutations (Appendix D.1).

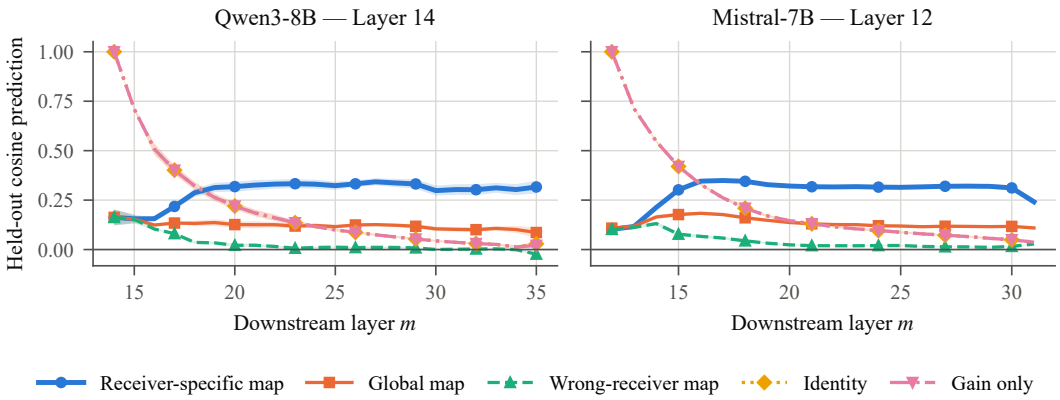


Figure 3: **A receiver-specific transformation predicts the footprints of unseen displacements.** Mean cosine between predicted and actual held-out footprints $c_{d,k}^{(m)}$ at every downstream layer m, on the 96×8 control panel. Maps are fit on 72 natural displacements and evaluated on 24 held-out displacements. Bands: ± 1 standard error across the 24 held-out displacements, after averaging the eight receivers within each. Identity and gain only coincide (a scalar gain does not change direction) and decay with depth; receiver-specific rank-16 maps keep signal through the final layer, global maps are weaker, and wrong-receiver maps fall to about zero.

Dose and horizon. On a random 256-pair subset of the ordinary matrices, the score stays informative up to the full transplant (Fisher–KL correlation 0.831 in Qwen3-8B and 0.907 in Mistral-7B at dose $\lambda = 1$; Figure 5). On a random 128-pair subset run on the true continuation, receiver-local sensitivity remains associated with KL summed over the next h teacher-forced positions, although the association weakens with the horizon ($0.857 \rightarrow 0.565$ in Qwen3-8B and $0.930 \rightarrow 0.812$ in Mistral-7B from $h = 1$ to 32; Appendix D.10). Teacher forcing does not establish persistence under unconstrained autoregressive generation.

Independent pretraining. On six independently pretrained Pythia-70M runs, all six partial Spearman correlations between $\log(1 + Q_{\mathrm{recv}})$ and transplant KL, controlling for update and state distance, nonlinear residual, and tangent alignment, are positive (median 0.800, 95% bootstrap interval $[0.742, 0.827]$; Appendix D.11). This uses ordinary transplants, not the crossed design: it shows that the local-geometry association replicates across independently pretrained small models, while the crossed-receiver evidence comes from the two large models.

6 Conclusion

A natural MLP displacement does not carry one context-free behavioral effect. Holding the update physically fixed while changing only its receiver produces large, systematic differences in exact next-token behavior, which an established receiver-local Fisher score ranks beyond generic receiver fragility and beyond scale, attribution, and alignment baselines. The receiver also rewrites the downstream direction of the effect, and receiver-specific maps recover a reproducible component of that transformation for unseen displacements. Analyses that move internal updates across contexts should treat the receiving state as part of the intervention.

Limitations. We study final-token MLP outputs and next-token behavior; other components and longer horizons are not covered. The large-scale evidence comes from one Qwen3-8B and one Mistral-7B checkpoint, and independent-pretraining replication exists only for Pythia-70M under ordinary transplants. The residual analysis is post hoc and descriptive. The receiver maps are linear and rank-16 and measure causal-effect geometry, not preservation of a semantic function. The crossed experiments use FP16 finite-difference Fisher scores; the FP32 suffix audit closes truncation error on the tested directions but does not make these scores numerically exact (Appendix D.12). Teacher-forced checks do not establish behavior under free generation. The study is diagnostic and proposes no steering method; preserving an intervention's intended causal effect across receiver states is a natural direction for future work (Appendix E).

References

Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, Usvsn Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. Pythia: A suite for analyzing large language models across training and scaling. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 2397–2430. PMLR, 23–29 Jul 2023.

Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems*, volume 34, pages 9574–9586. Curran Associates, Inc., 2021.

Nicholas Goldowsky-Dill, Chris MacLeod, Lucas Sato, and Aryaman Arora. Localizing model behavior with path patching. *arXiv preprint arXiv:2304.05969*, 2023. URL https://arxiv.org/abs/2304.05969.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023.

János Kramár, Tom Lieberum, Rohin Shah, and Neel Nanda. AtP*: An efficient and scalable method for localizing LLM behaviour to components. *arXiv preprint arXiv:2403.00745*, 2024. URL https://arxiv.org/abs/2403.00745.

James Martens. New insights and perspectives on the natural gradient method. *Journal of Machine Learning Research*, 21(146):1–76, 2020.

Kiho Park, Todd Nief, Yo Joong Choe, and Victor Veitch. The information geometry of softmax: Probing and steering. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL https://arxiv.org/abs/2602.15293.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

Eric Todd, Millicent L. Li, Arnab Sen Sharma, Aaron Mueller, Byron C. Wallace, and David Bau. Function vectors in large language models. In *The Twelfth International Conference on Learning Representations*, 2024.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023. URL https://arxiv.org/abs/2308.10248.

Oskar van der Wal, Pietro Lesci, Max Müller-Eberstein, Naomi Saphra, Hailey Schoelkopf, Willem Zuidema, and Stella Biderman. PolyPythias: Stability and outliers across fifty language model pre-training runs. In *The Thirteenth International Conference on Learning Representations*, 2025.

Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart M. Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, pages 12388–12401. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper/2020/hash/92650b2e92217715fe312e6fa7b90d82-Abstract.html.

Sihan Wang and Jiayi Zhao. FishBack: Pullback fisher geometry for optimal activation steering in transformers. *arXiv preprint arXiv:2605.17231*, 2026. URL https://arxiv.org/abs/2605.17231.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *The Twelfth International Conference on Learning Representations*, 2024.

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crossed-receiver stages—four bucket-1 depths for each model plus bucket-2 and bucket-3 primary-depth replications—for 131,072 physical crossed-receiver interventions in total, together with a 65,280-pair Mistral full-core stage. Instrumented successful-stage runtimes in `ALL_STAGE_SUMMARIES.csv` sum to 631.3 minutes (10.52 hours wall-clock): 437.8 minutes for the crossed-receiver stages and 193.4 minutes for the Mistral full-core stage. Because both T4s shard each large model, this corresponds to roughly 21 T4 GPU-hours if both devices are counted for the full measured interval. The observed end-to-end successful notebook run was approximately 10–12 hours, including model loading, native-state caching, and ordinary notebook/session overhead. The eight ordinary 64-context depth targets ran on the same hardware class and total 134.9 minutes of recorded target runtime; the two 96×8 control-panel runs used the same hardware class, but their wall-clock time was not instrumented, so we do not report it. These figures exclude exploratory development and failed engineering attempts, which used additional compute. The saved row-level artifact can be reanalyzed on CPU without rerunning the GPU experiments (Appendix C.6). The exact FP32 downstream-suffix JVP audit (Appendix D.12.2; 16 pairs per model) also ran on Kaggle with two NVIDIA T4 GPUs (16 GB each): natural states were produced with the FP16 model, and the downstream suffix was evaluated in FP32 with TF32 disabled, split across both GPUs. Its runtime was not recorded, so we do not report it.

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